

Lymphoid Neoplasms

Natasha M. Savage, MD, FCAP • built to follow alongside the 70-slide lecture deck

Official Week 7 LOs

Slide-order companion

4 ScholarRx Bricks

Morphology + marker maps

Official-LO gap fills labeled

Factoid sheet companion

Overview: Official Week 7 Learning Objectives

LO 1. Define lymphoma and compare with leukemia.

LO 2. Discuss the methods and ancillary studies used in diagnosis of lymphoid neoplasms, including IHC, flow cytometry, and FISH.

LO 3. Discuss the important clinicopathologic and genetic features of acute lymphoblastic leukemia/lymphoma, including B-ALL/LBL and T-ALL/LBL.

LO 4. Discuss the major subtypes of Hodgkin lymphoma, including clinical presentation, morphology, immunophenotype, treatment, and clinical course: classic HL and NLPHL.

LO 5. Discuss the important clinicopathologic and genetic features of common NHL subtypes: CLL/SLL, hairy cell leukemia, follicular lymphoma, MALT lymphoma, mantle cell lymphoma, DLBCL, Burkitt lymphoma, and mycosis fungoides/Sézary syndrome.

LO 6. Discuss plasma-cell neoplasms, including MGUS, smoldering myeloma, plasma-cell myeloma, and primary amyloidosis (AL).

SCHOLARRX BRICKS PAIRED WITH THIS LECTURE

Hematopoietic Malignancies: Foundations & Frameworks

Chronic Lymphoproliferative Disorders: Foundations & Frameworks

Lymphoma: Foundations & Frameworks

Hematopoietic Malignancies: Putting It All Together

LO 1

Slides 4-8: leukemia vs lymphoma and immature vs mature B/T/NK organization.

LO 2

Companion-supported: morphology + IHC + flow + FISH/genetics + clinical context as an integrated diagnosis.

LO 3

Slides 9-16: B-ALL/LBL vs T-ALL/LBL, TdT, childhood epidemiology, marrow vs mediastinal presentation, prognostic genetics.

LO 4

Slides 17-29 + gap-fill: classic HL in depth; NLPHL supplied at recognition level from the companion because the deck explicitly does not cover it in detail.

LO 5

Slides 30-60 + gap-fill: mature B-cell NHL patterns, MF/Sézary, and hairy cell leukemia from the companion because HCL is in the official LO but absent from the 70-slide deck.

LO 6

Slides 62-69 + enrichment: paraprotein, MGUS/smoldering/myeloma spectrum, CRAB, morphology, and AL amyloid.

SOURCE NOTE - IMPORTANT

The **70-slide Savage deck** is the primary source and slide-order anchor. The uploaded companion calls itself “Lectures 2 and 3” and says it was built around a **114-slide** version of the lecture. Because it contains material that is missing from this current deck, I use it only to fill official learning-objective gaps and clearly label those additions rather than pretending both sources are identical.

THE LECTURE'S MASTER STRATEGY

Do not memorize lymphomas as a random alphabet soup. For every neoplasm, ask:

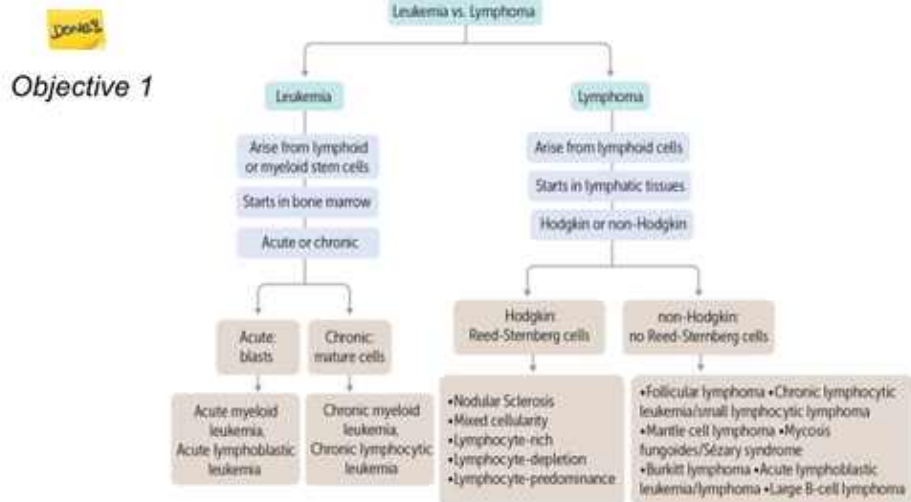
1) What cell/stage is malignant? 2) Where does it present? 3) What does it look like? 4) What markers does it express? 5) What genetic lesion drives it? 6) Is it indolent or aggressive?

Slides 4-8

Leukemia vs lymphoma and the family tree

LO 1

LEUKEMIA VERSUS LYMPHOMA



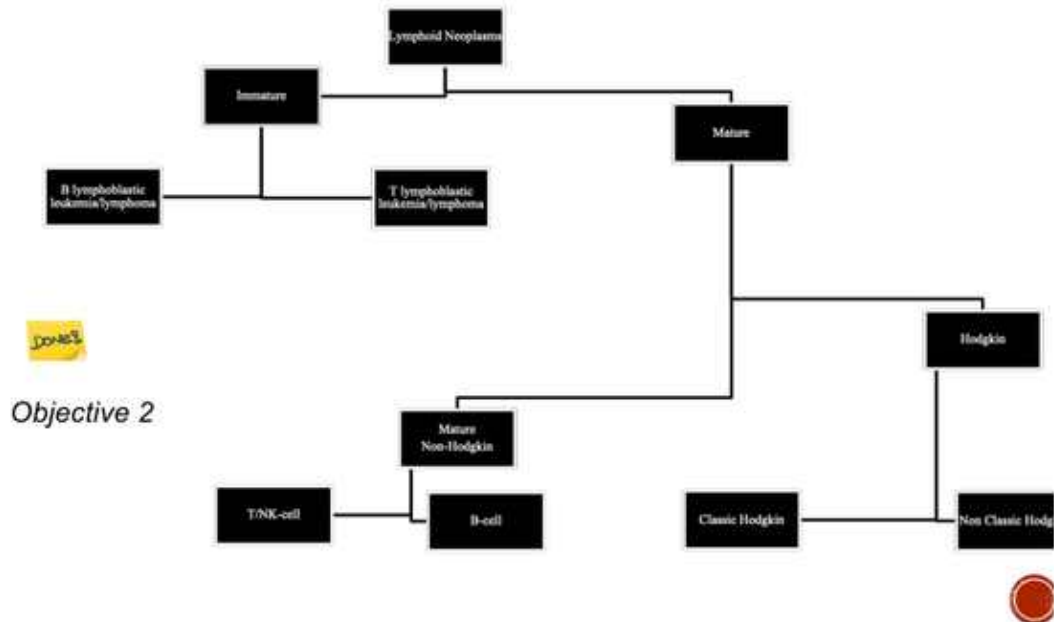
Lecture slide 5 - full slide retained. Savage organizes the distinction by primary site and presentation rather than pretending leukemia and lymphoma are always different biological diseases.

	Leukemia	Lymphoma
Primary presentation	Bone marrow / peripheral blood	Discrete tissue or lymph-node mass
Spread	Circulating cells are common	May disseminate secondarily into blood/ marrow
Key conceptual point	The terms can describe presentation of the same biologic entity. CLL/ SLL and ALL/LBL are the classic examples.	

NEED TO MEMORIZE

**CLL and SLL = same malignant B cell, different dominant compartment.
ALL and LBL = same precursor disease spectrum, different dominant compartment.**

LYMPHOID NEOPLASMS



Objective 2

Lecture slide 7 - full slide retained. The deck first separates immature precursor B/T neoplasms from mature lymphoid disease, then mature disease into Hodgkin and non-Hodgkin branches.

WHY DEVELOPMENTAL STAGE MAKES THE REST EASIER

Markers and behavior reflect what the normal counterpart should be doing. **TdT/CD34** point toward immaturity. Mature B-cell neoplasms carry B-cell markers and may retain stage-specific programs such as germinal-center CD10 expression or mantle-zone cyclin D1 biology.

Official LO 2 companion

How lymphoid neoplasms are actually diagnosed

LO 2

WHY THIS SECTION IS HERE

The official Week 7 objectives explicitly require **IHC, flow cytometry, and FISH**. The current deck demonstrates many immunostains and cytogenetic lesions but does not give a single diagnostic-toolbox slide. The uploaded companion supplies the framework below.

Tool	What it answers	Best use
Morphology	What do the cells and tissue architecture look like?	Blast vs mature cell; follicular vs diffuse growth; RS cells; starry sky; plasma-cell morphology.
IHC	Where are proteins expressed in intact tissue?	Preserves architecture; e.g., CD20, CD3, CD15/CD30, cyclin D1, BCL2.
Flow cytometry	What markers are co-expressed on individual suspended cells?	Rapid immunophenotyping and clonality; especially blood, marrow, and fresh tissue.
FISH / cytogenetics	What rearrangement or copy-number lesion is present?	t(14;18), t(11;14), MYC rearrangement, BCR-ABL1, etc.
Clinical context	Who has the disease and where is it?	Age, site, tempo, B symptoms, marrow/blood involvement, extranodal pattern.

CLONALITY IDEA

A reactive B-cell population is **polyclonal**; a neoplastic population is often **light-chain restricted** (predominantly kappa or lambda), supporting derivation from one clone.

EXAM TRAP

No single marker diagnoses most lymphomas. CD5, for example, is normally a T-cell marker but is aberrantly expressed in both CLL/SLL and mantle-cell lymphoma. You need a **pattern**, not one stain.

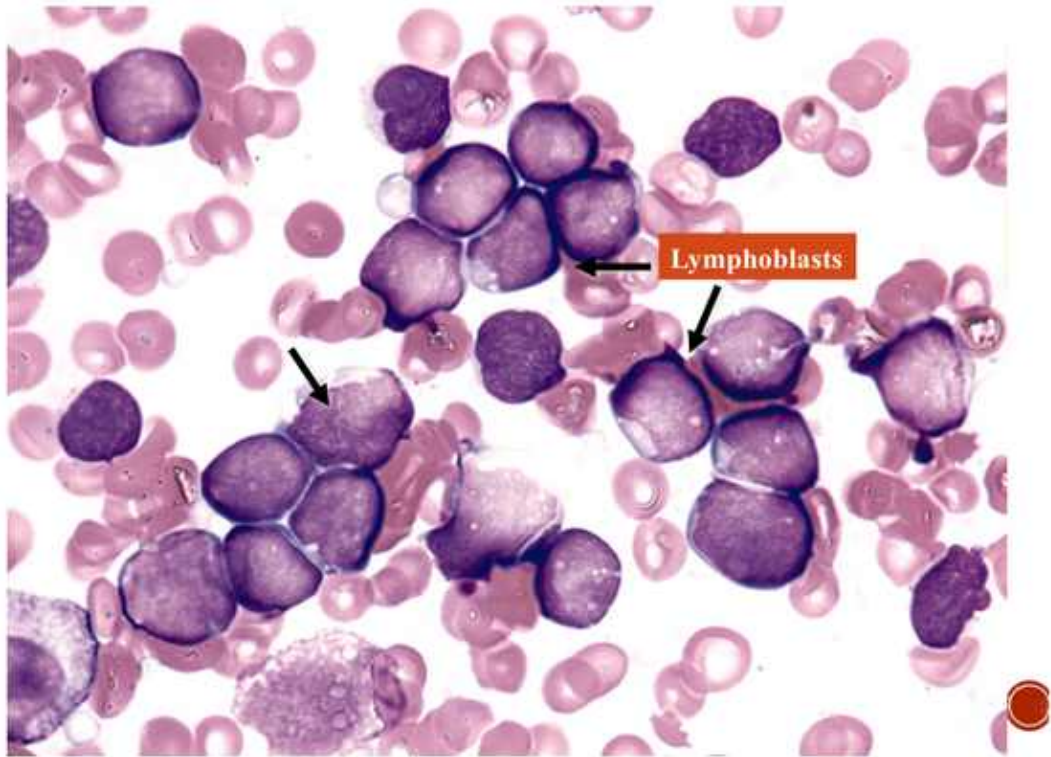
Slides 9-10: same precursor disease, different dominant presentation

- ALL/LBL arises from **immature B or T lymphoid precursors**.
- It is predominantly a disease of childhood; Savage gives a peak around **3-7 years**.
- **B-ALL/LBL** is more common and more often presents with marrow/blood disease.
- **T-ALL/LBL** more often presents as a tissue mass, especially a **mediastinal/thymic mass in adolescent males**.

Slides 11-12: marrow failure + blast phenotype

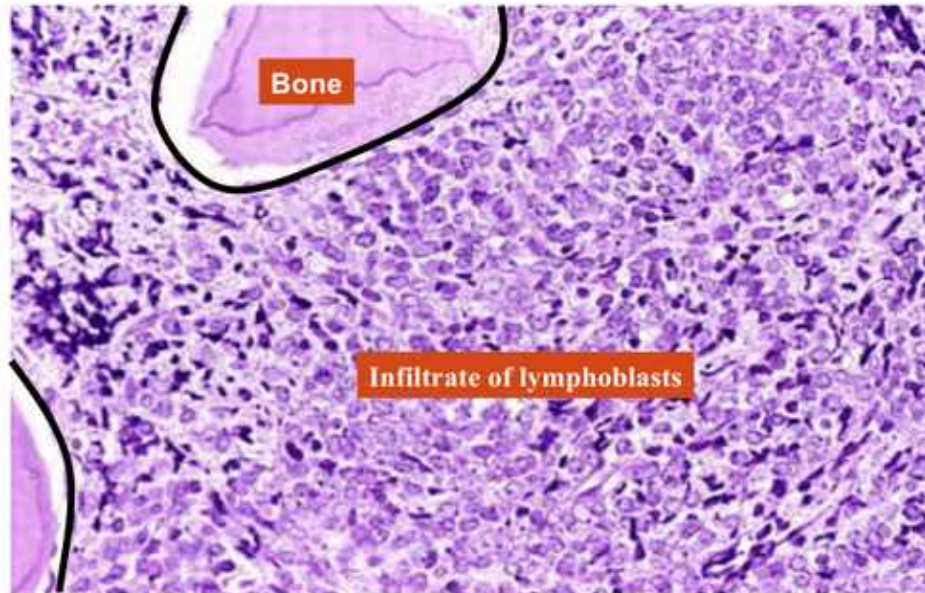
- Marrow replacement -> **anemia, thrombocytopenia, neutropenia/ infections, bone pain**.
- B-lymphoblasts: small-medium blasts with scant cytoplasm and **no Auer rods**.
- Slide phenotype: **CD19+, CD20 +/- , CD10+, TdT+, CD34+**.

B-ALL: BLOOD



Lecture slide 13 - full slide retained. Use the full-slide smear to recognize a blast-rich picture: large nuclei, high nuclear-to-cytoplasmic ratio, scant basophilic cytoplasm, and immature chromatin.

B-ALL: BONE MARROW



Lecture slide 14 - full slide retained. The marrow is replaced by lymphoblasts, explaining cytopenias and bone pain.

NEED TO MEMORIZE - IMMATURITY

TdT + CD34 = precursor/blast program.

B-ALL: CD19/CD10. T-ALL: T-lineage markers such as CD3/CD7 plus TdT.

Slides 15-16: prognosis and B vs T

- Favorable in the deck: age 2-10, lower leukocyte count, **t(12;21)**, rapid initial response.
- Unfavorable in the deck: **Philadelphia chromosome t(9;22)**, meningeal/testicular involvement.
- T-ALL/LBL: adolescent male, mediastinal mass, CD2/CD3/CD7/TdT.

BOARD ENRICHMENT - SANCTUARY SITES

CNS and testes are classic sanctuary sites in ALL, which is why CNS-directed prophylaxis is built into treatment strategies. This comes from the uploaded companion rather than the current 70-slide deck.

LO 3 checkpoint

Can you distinguish B-ALL from T-ALL using **age/site + TdT + lineage markers**, and explain why marrow replacement causes the presenting symptoms?

Slides 17-29

Hodgkin lymphoma: Reed-Sternberg biology, phenotype, spread, staging

LO
4

Slides 19-21: what makes Hodgkin lymphoma clinically distinctive

- Primarily **lymph-node based**.
- Usually presents **above the diaphragm**.
- Classically spreads in a **predictable contiguous** fashion through nodal groups.
- B symptoms: fever, night sweats/chills, weight loss; pruritus may occur.
- Diagnosis requires tissue biopsy.

Slides 22-24: the malignant cell is rare but memorable

Classic Hodgkin lymphoma is a clonal B-cell neoplasm in which the malignant Hodgkin/Reed-Sternberg cells are surrounded by a large reactive inflammatory background.

REED-STERNBERG CELL

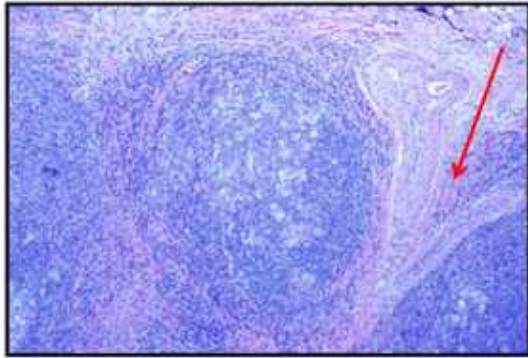


Lecture slide 23 - full slide retained. Classic RS morphology: very large cell with two nuclear lobes/nuclei and prominent eosinophilic nucleoli with perinuclear clearing - the classic 'owl-eye' concept.

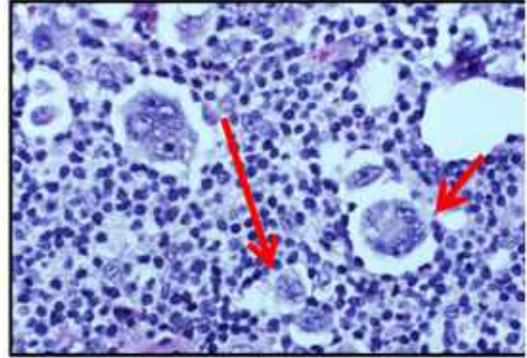
CLASSIC HODGKIN PHENOTYPE

CD15+ • CD30+ • CD45- • usually CD20- / weak B-cell program.

NODULAR SCLEROSIS CHL: MORPHOLOGY



Nodules with dense collagen bands

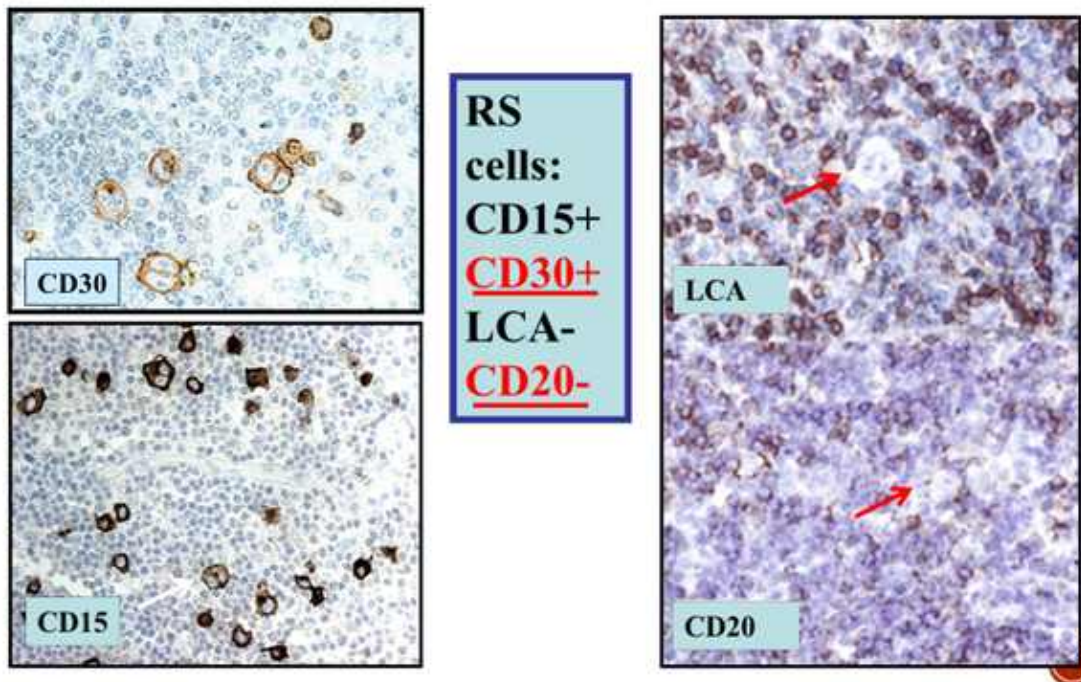


Neoplastic "Lacunar cells" seen



Lecture slide 25 - full slide retained. Nodular sclerosis shows broad collagen bands dividing the node and characteristic lacunar variants of RS cells; it commonly presents with a mediastinal mass.

CHL: IMMUNOPHENOTYPE

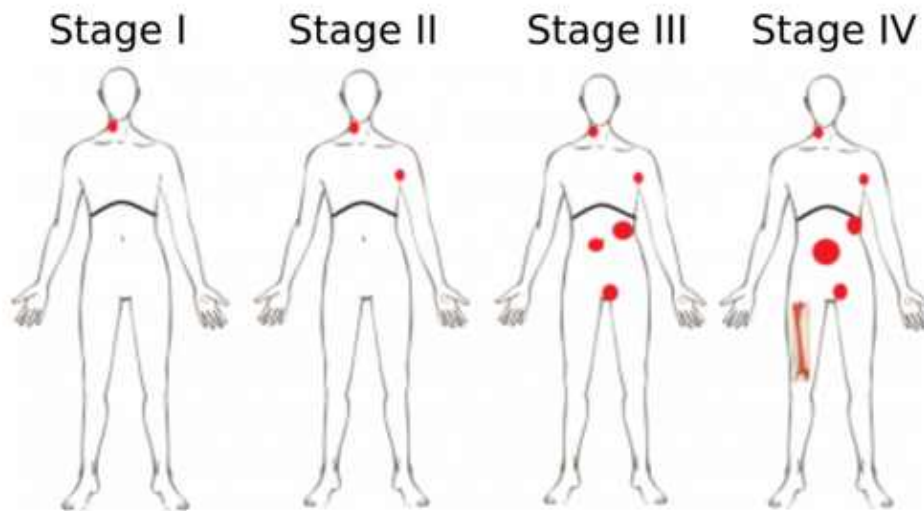


Lecture slide 28 - full slide retained. The immunostain panel visually reinforces CD15/CD30 positivity and weak/absent conventional leukocyte/B-cell-marker expression.

HODGKIN VS NHL SPREAD

Hodgkin: more orderly contiguous nodal spread. **NHL:** often multicentric/disseminated at diagnosis with extranodal disease more common.

ANN ARBOR STAGING SYSTEM FOR HL



A: no symptoms

B: + B symptoms (i.e., fever, chills, night sweats, unintentional weight loss)

E: extranodal extension

X: bulky disease (>10cm)

don't

Objective 5



Lecture slide 29 - full slide retained. Stage I through IV is based on distribution; the slide also labels A vs B symptoms, E for extranodal extension, and X for bulky disease.

OFFICIAL LO GAP-FILL - NLPHL

The deck explicitly says nodular lymphocyte-predominant Hodgkin lymphoma will not be discussed in detail, but the official LO requires it. Recognition-level companion facts:

- Characteristic **LP / “popcorn” cells**.
- **CD20+ and CD45+**, unlike classic HL.
- **CD15- and CD30-**, essentially the opposite of classic RS immunophenotype.
- Typically more indolent and can have late recurrences.

LO 4 checkpoint

Can you identify classic HL from a node biopsy, state CD15/CD30/CD45, explain contiguous spread, and distinguish NLPHL's popcorn-cell/CD20+ pattern?

Slides 30-55

Mature B-cell non-Hodgkin lymphomas

LO 5

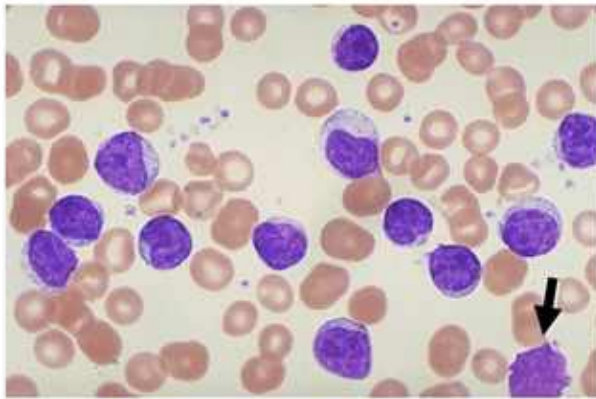
Slides 31-34: behavior matters as much as name

- NHLs are a heterogeneous group of mature B, T, or NK-cell neoplasms.
- They are often **multicentric and disseminated** at diagnosis and can arise extranodally.
- Indolent diseases may be widespread but slow; aggressive lymphomas grow fast but can be more treatment-responsive/curable.
- The deck's B-cell split: lower-grade CLL/SLL, follicular, MALT; mantle-cell "in a class by itself"; high-grade DLBCL and Burkitt.

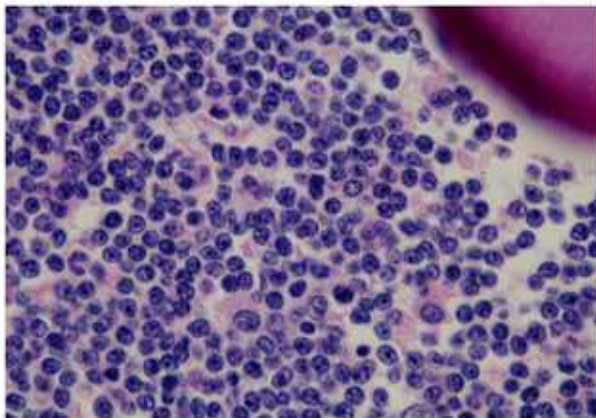
CLL / SLL

Slides 35-38

- Same underlying B-cell neoplasm; CLL = blood/marrow dominant, SLL = node/tissue dominant.
- Median age around 70; often indolent.
- Can cause lymphadenopathy, splenomegaly, B symptoms, **autoimmune hemolytic anemia/ITP**, and hypogammaglobulinemia.
- Immunophenotype: **CD19+**, **dim CD20+**, **aberrant CD5+**, **CD23+**.



Lymphocytosis with small mature lymphocytes & smudge cell (arrow)



Bone marrow involvement by CLL



Lecture slide 38 - full slide retained. The slide combines the classic peripheral-blood pattern of small mature lymphocytes and a smudge cell with marrow involvement.

CLL VS MANTLE CELL

Both can be CD5+ B-cell neoplasms.

CLL/SLL = **CD23+**. Mantle = **CD23-** + **cyclin D1+**.

COMPANION ENRICHMENT - RICHTER TRANSFORMATION

A previously indolent CLL/SLL patient who suddenly develops a rapidly enlarging node/mass, rising LDH, or new systemic symptoms should raise concern for transformation to an aggressive lymphoma, classically DLBCL.

Hairy Cell Leukemia - official-LO gap-fill

NOT DEVELOPED IN THE CURRENT 70-SLIDE DECK

The official LO lists HCL, so the uploaded companion supplies the recognition set:

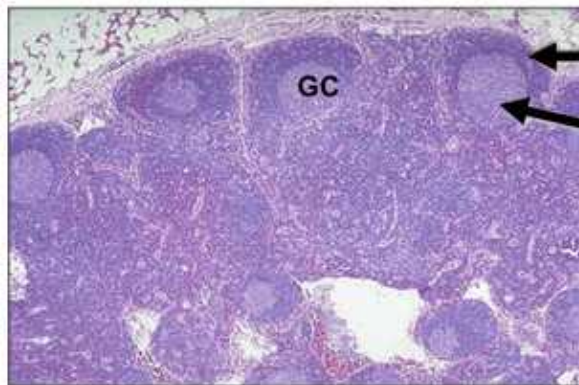
- Mature B cells with fine **hair-like cytoplasmic projections**.
- **Massive splenomegaly + pancytopenia + dry tap** from marrow fibrosis.
- Markers: CD19/CD20, CD11c, CD25, CD103, Annexin A1; classic **TRAP+**.
- **BRAF V600E** is a signature driver.
- Classic treatment association: **cladribine**.

Follicular Lymphoma

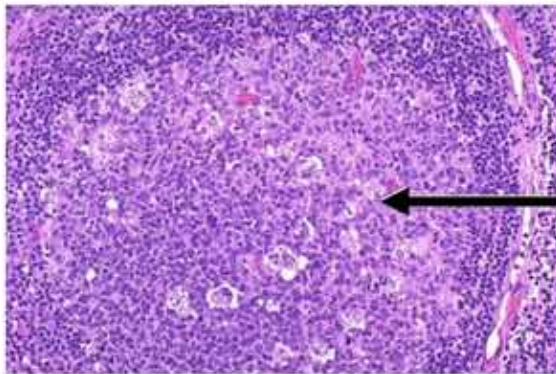
Slides 39-43

- Indolent germinal-center B-cell lymphoma with painless generalized lymphadenopathy.
- Can transform to DLBCL.
- Node shows **uniform malignant follicles** lacking normal germinal-center polarity, mantle zones, and tingible-body macrophages.
- CD19/CD20/CD10+, with **t(14;18) IgH-BCL2** -> BCL2 overexpression -> impaired apoptosis.

Reactive lymph node



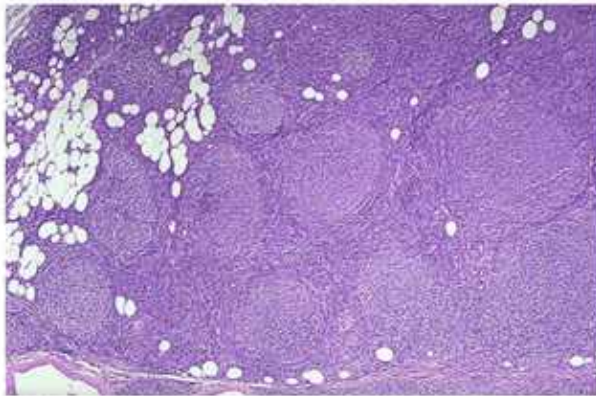
Preserved mantle zone
Reactive germinal center
Follicles of varying sizes



Tingible body macrophage

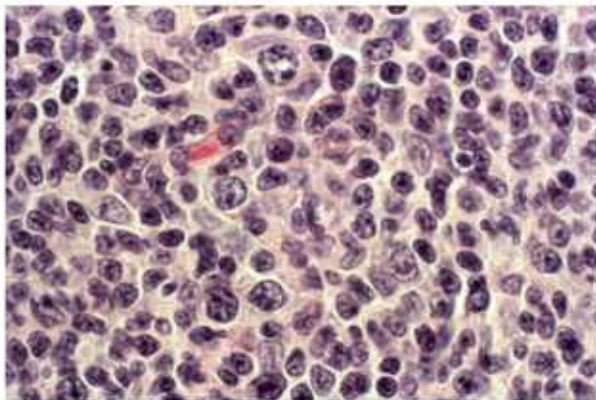


Lecture slide 41 - full slide retained. Reactive follicles vary in size, preserve mantle zones/polarity, and contain tingible-body macrophages because normal germinal-center B cells undergo apoptosis.



Follicular lymphoma

Uniform follicles
Lack of mantle zone
Lack of polarity



No tingible body macrophages



Lecture slide 42 - full slide retained. Malignant follicles are more monotonous and lack mantle zones, polarity, and tingible-body macrophages - a direct visual contrast with the preceding reactive node.

FOLLICULAR LYMPHOMA

t(14;18) -> BCL2 -> blocks apoptosis.

Indolent, waxing/waning, can transform to DLBCL.

MALT / Marginal Zone Lymphoma

Slide 44

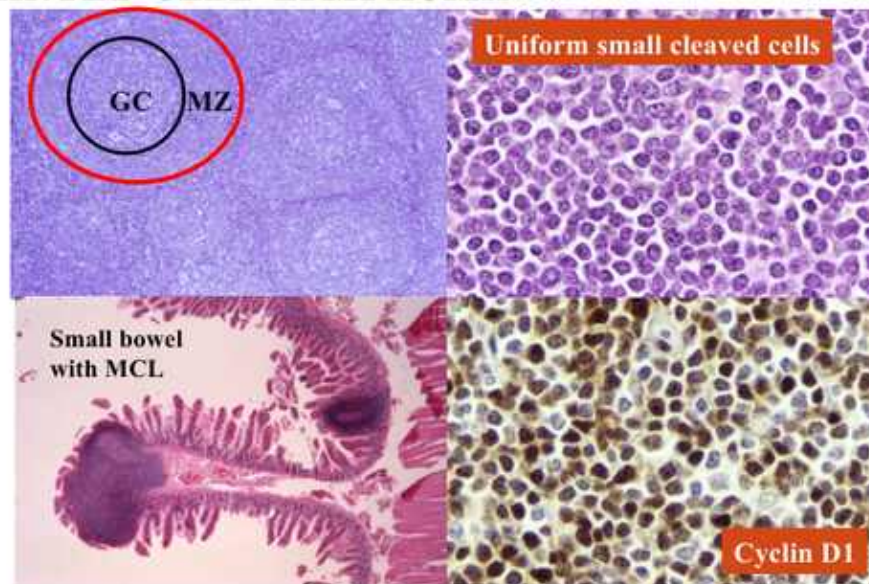
- Indolent B-cell lymphoma associated with **chronic antigenic/inflammatory stimulation.**
- Examples in the deck: **H. pylori stomach, Sjögren salivary gland, Hashimoto thyroiditis.**
- Localized gastric MALT may regress after **H. pylori eradication.**

Mantle Cell Lymphoma

Slides 45-48

- Older males; nodes, marrow, GI tract/Waldeyer ring can be involved.
- More aggressive than the classic low-grade small-cell lymphomas.
- CD19/CD20+ with **CD5+**, but unlike CLL it is **CD23-**.
- **t(11;14) IgH-CCND1** -> **cyclin D1 overexpression** -> cell-cycle drive.

MANTLE CELL LYMPHOMA



GC: Germinal center

MZ: Mantle zone



Lecture slide 48 - full slide retained. The slide shows mantle-zone growth, small irregular/cleaved cells, GI involvement, and nuclear cyclin D1 staining.

TRANSLOCATION TRIO

Follicular = 14;18 = BCL2.

Mantle = 11;14 = cyclin D1.

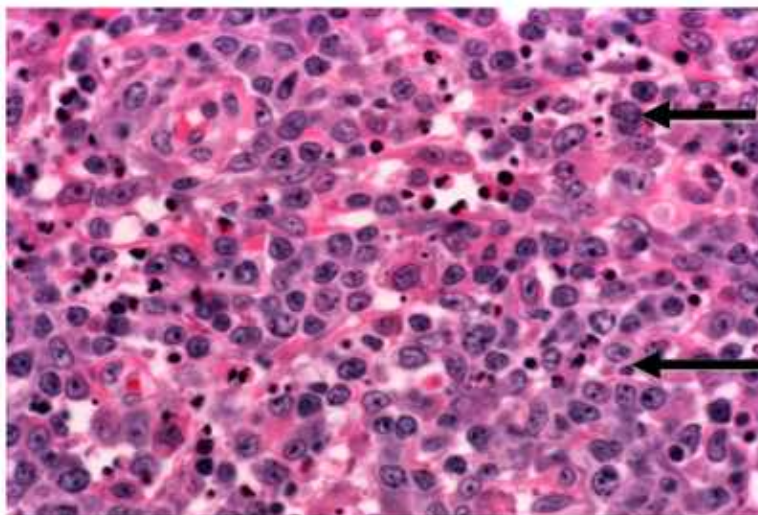
Burkitt = 8;14 = MYC.

Diffuse Large B-Cell Lymphoma

Slides 49-51

- **Most common lymphoma in the US** in the deck.
- Usually older adults; can arise de novo or by transformation from an indolent lymphoma.
- Rapidly enlarging symptomatic nodal or extranodal mass.
- Aggressive without treatment, but durable remissions are possible.

DIFFUSE LARGE B-CELL LYMPHOMA: MORPHOLOGY



Large cell lymphoma
cell >2x size of
normal lymphocyte

Normal
lymphocyte

Lecture slide 51 - full slide retained. The defining visual idea is size: malignant large cells are more than twice the size of a normal small lymphocyte.

CURRENT TREATMENT ENRICHMENT - DLBCL

R-CHOP remains the classic board regimen association. Current U.S. therapy has evolved: FDA also approved **polatuzumab vedotin + R-CHP** for selected previously untreated DLBCL/high-grade B-cell lymphoma patients. Keep this as current clinical enrichment rather than replacing the lecture's foundational behavior/morphology focus.

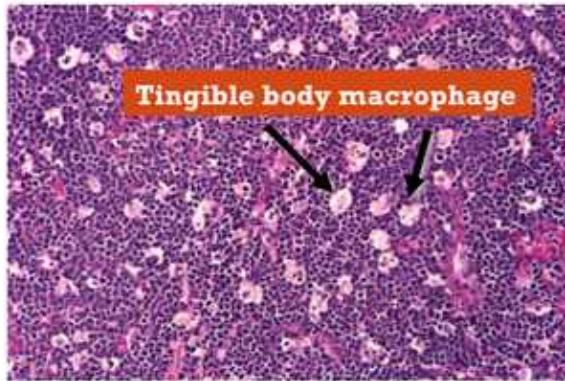
FDA polatuzumab + R-CHP approval

Burkitt Lymphoma

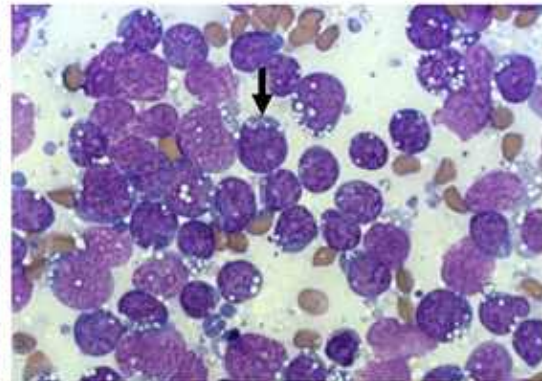
Slides 52-54

- **Extremely aggressive / STAT diagnosis**; high tumor growth fraction.
- Endemic African form often involves jaw/maxilla; sporadic North American form often abdominal; immunodeficiency-associated form occurs with HIV.
- CD19/CD20/CD22/CD10 and surface Ig.
- **t(8;14) MYC-IgH.**
- “Starry sky” = pale tingible-body macrophages among rapidly proliferating/dying tumor cells.

BURKITT LYMPHOMA



Starry-sky pattern



Cytoplasmic vacuoles



Lecture slide 54 - full slide retained. The full slide shows the starry-sky pattern, cytoplasmic vacuoles, and tingible-body macrophages that reflect Burkitt's extraordinarily high proliferation and turnover.

BURKITT COLD SET

MYC • t(8;14) • CD10+ mature B cell • starry sky • very high Ki-67 • jaw or abdominal mass depending epidemiologic form.

Slide 55: rituximab

Rituximab is an anti-**CD20** monoclonal antibody and was a major therapeutic shift for mature B-cell malignancies. Mechanistically, antibody binding enables immune-mediated killing of CD20+ B cells.

LO 5 B-cell checkpoint

Can you identify CLL/SLL, HCL, follicular, MALT, mantle, DLBCL, and Burkitt from a **one-line combination of age/site + morphology + marker + genetic lesion + tempo?**

Slides 56-60

Mature T-cell lymphoma: mycosis fungoides / Sézary syndrome

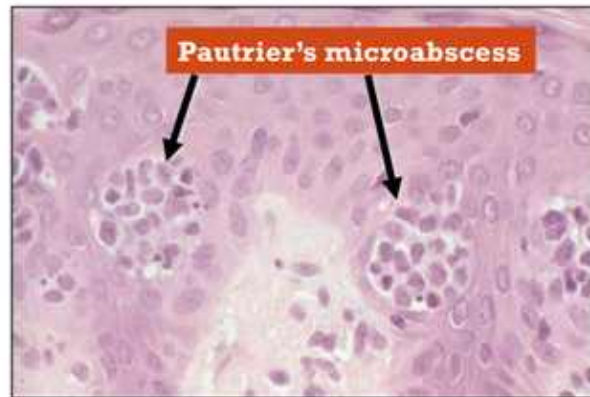
LO 5

Slides 57-60

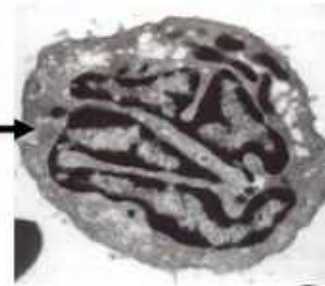
- Cutaneous T-cell lymphoma, usually adults/elderly.
- Can mimic eczema, psoriasis, or dermatitis for years.
- Patches -> plaques -> tumors; classically non-sun-exposed skin.
- Histology: **epidermotropism + Pautrier microabscesses**.
- Neoplastic cells have **cerebriform nuclei**; Sézary syndrome is the leukemic/erythrodermic form.
- Slide phenotype: **CD3+, CD4+, loss of CD7**.

MF

- Epidermotropic infiltrate
- Pautrier microabscess
- Cerebriform nuclei (Sézary cell)
- CD3+, CD7-, CD4+ (Helper T-cells)



Electron
microscopy
revealing
cerebriform
nucleus



Lecture slide 60 - full slide retained. The slide shows epidermotropism/Pautrier microabscess and a cerebriform Sézary nucleus - the two visual findings to remember.

NEED TO MEMORIZE

Mycosis fungoides = skin patches/plaques + epidermotropic CD4 T cells + Pautrier microabscesses.

Sézary = erythroderma + circulating cerebriform malignant T cells.

Slides 62-69

Plasma-cell neoplasms: MGUS -> smoldering -> myeloma + AL amyloid

LO
6

Slides 62-64: think spectrum, not one disease

- Clonal plasma cells produce a **monoclonal immunoglobulin/paraprotein**.
- Serum protein electrophoresis may show an **M-spike**.
- Suppression of normal immunoglobulin diversity contributes to infection risk.
- The deck places **MGUS -> smoldering myeloma -> symptomatic myeloma** on a progression spectrum.
- Deck teaching point: roughly **1% of MGUS cases per year progress**; smoldering myeloma has greater progression risk.

Slides 65-66: why myeloma causes systemic disease

- Marrow replacement -> **anemia and other cytopenias**.
- Osteoclast-driving marrow biology -> **lytic lesions, bone pain, fractures, hypercalcemia**.
- Monoclonal light chains/proteins -> **renal injury** and can form AL amyloid.
- Reduced normal Ig -> infections.

CRAB

Calcium high • Renal injury • Anemia • Bone lesions.



Rouleaux formation in blood:
Cells stick to each other like stacks of coins



Lecture slide 67 - full slide retained. High circulating protein reduces normal RBC repulsion, producing the classic stacked-coin rouleaux pattern.

MALIGNANT PLASMA CELLS

- Morphology
 - Ranging from mature plasma cells to markedly atypical pleomorphic cells
- Immunophenotype
 - Positive for CD138 (high density) often with aberrant CD56
 - Clonal cytoplasmic immunoglobulin (Ig)



Lecture slide 68 - full slide retained. Malignant plasma cells retain eccentric nuclei and abundant cytoplasm; the slide emphasizes CD138 and clonal cytoplasmic immunoglobulin, often with aberrant CD56.

OFFICIAL LO GAP-FILL - MGUS, SMOLDERING MYELOMA, AL AMYLOID

- **MGUS:** monoclonal protein with low clonal burden and **no myeloma-defining organ damage**; companion uses <10% marrow plasma cells as the classic board threshold.
- **Smoldering myeloma:** higher clonal burden than MGUS but no myeloma-defining event.
- **AL amyloidosis:** misfolded monoclonal immunoglobulin light chains deposit in tissues; classic board clues include nephrotic proteinuria, restrictive cardiomyopathy, macroglossia, periorbital purpura; Congo red shows apple-green birefringence.

CURRENT DIAGNOSTIC ENRICHMENT - SLIM-CRAB

Modern IMWG criteria allow active myeloma to be diagnosed before classic CRAB damage if certain high-risk myeloma-defining biomarkers are present. The **SLiM** additions are: **$\geq 60\%$ clonal marrow plasma cells, involved:uninvolved serum free-light-chain ratio ≥ 100 (with involved FLC ≥ 100 mg/L), or ≥ 2 MRI focal lesions.** This is current external enrichment beyond the lecture's CRAB-focused slides.

IMWG/International Myeloma Foundation diagnostic criteria

LO 6 checkpoint

Can you explain the difference between **having an M-spike** and having **active myeloma**, and connect plasma-cell biology to CRAB, rouleaux, and AL amyloid?

Synthesis

One-table lymphoma fingerprint map

LO 3-6

Disease	Cell / markers	Genetic / visual clue	Tempo / presentation
B-ALL/LBL	TdT+, CD34+, CD19+, CD10+	Blasts; t(12;21) favorable, t(9;22) unfavorable in deck	Child; marrow/blood dominant
T-ALL/LBL	TdT+, CD3/ CD7+	Precursor T cell	Adolescent male + mediastinal mass
Classic HL	RS cell; CD15+, CD30+, CD45-	Owl-eye RS; nodular sclerosis has lacunar cells/bands	Contiguous spread; often above diaphragm
NLPHL	CD20+, CD45+, CD15-/CD30-	Popcorn/L&H cells	Indolent, late recurrences possible
CLL/SLL	CD5+, CD23+, dim CD20	Smudge cells	Older; indolent; AIHA/ hypogammaglobulinemia
Hairy cell	CD11c/CD25/ CD103; TRAP	BRAF V600E; hairy cells	Splenomegaly + pancytopenia + dry tap
Follicular	CD10+, B-cell markers	t(14;18) BCL2; uniform follicles	Indolent, waxing/waning; can transform
MALT	Mature B-cell	Chronic antigenic stimulation	H. pylori/Sjögren/Hashimoto associations
Mantle	CD5+, CD23-, cyclin D1+	t(11;14) CCND1	Older male; more aggressive; GI can be involved
DLBCL	Mature large B cells	Diffuse large cells; variable BCL2/BCL6/ MYC biology	Rapidly growing mass; aggressive but potentially curable
Burkitt	CD10+, CD20+, surface Ig	t(8;14) MYC; starry sky	Extremely rapid; jaw/abdomen depending form
MF / Sézary	CD3+, CD4+, CD7 loss	Pautrier microabscess / cerebriform cell	Skin-limited -> erythrodermic/ leukemic spectrum
Plasma-cell myeloma	CD138+, clonal Ig	M-spike, rouleaux, lytic disease	CRAB / SLiM-CRAB myeloma- defining events

THE THREE TRANSLOCATIONS YOU ABSOLUTELY CANNOT MIX UP

14;18 = Follicular = BCL2.

11;14 = Mantle = Cyclin D1.

8;14 = Burkitt = MYC.

COMMON EXAM TRAPS

- **CD5+ B cell** does not uniquely mean CLL: CLL is CD23+, mantle is CD23- / cyclin D1+.
- **TdT** = immaturity, not T-cell lineage specifically.
- **Starry sky** reflects very high turnover and is classically Burkitt, but morphology must be integrated with phenotype/genetics.
- **RS-like cells** are not enough by themselves: classic Hodgkin diagnosis is architectural + morphologic + immunophenotypic.
- An **M-spike alone** does not equal multiple myeloma.

Learning-Objective Synthesis

Official LO	What you should be able to do without notes
LO 1	Explain leukemia vs lymphoma as presentation/site terminology and give CLL/SLL + ALL/LBL as overlap examples.
LO 2	Explain how morphology, IHC, flow cytometry, FISH/genetics, clonality, and clinical context converge on a diagnosis.
LO 3	Recognize precursor B/T disease using TdT and lineage markers, age/site, marrow failure, and prognostic cytogenetics.
LO 4	Distinguish classic HL from NLPHL and identify RS cells, CD15/CD30, nodular sclerosis, contiguous spread, and staging logic.
LO 5	Recognize every listed common NHL by its fingerprint: CLL/SLL, HCL, FL, MALT, MCL, DLBCL, Burkitt, MF/SS.
LO 6	Distinguish MGUS, smoldering disease, active myeloma, and AL amyloidosis; explain CRAB and paraprotein consequences.

LAST 60 SECONDS

ALL: TdT; B = child/marrow, T = teen/mediastinum.

Classic HL: RS cell, CD15/CD30, contiguous spread.

CLL: CD5/CD23 + smudge. **Mantle:** CD5/CD23- + cyclin D1, 11;14.

Follicular: 14;18 BCL2. **Burkitt:** 8;14 MYC + starry sky.

DLBCL: most common NHL, rapidly enlarging large-cell mass.

MF/Sézary: Pautrier / cerebriform cells.

Myeloma: monoclonal plasma cells + CRAB/SLiM-CRAB, M-spike, rouleaux, lytic lesions.

Source hierarchy: Savage's current 70-slide Lymphoid Neoplasms deck is the primary slide-order source. The official Week 7 objective sheet controls emphasis. The uploaded 114-slide-derived companion is used only for official-LO gaps such as diagnostic ancillary studies, hairy cell leukemia, NLP HL, sanctuary-site context, Richter transformation, and additional plasma-cell distinctions. Current DLBCL and myeloma criteria are explicitly labeled external enrichment.